

Sourav Danyal

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SUMMARY

Motivated Data Science and AI enthusiast with hands-on experience in building ML models, RAG applications, and API services. Passionate about applying data-driven approaches to solve real-world problems.

EDUCATION

Central University of Himachal Pradesh <i>Master of Computer Applications (MCA)</i>	2024 – Present
Government Degree College Shahpur (HPU) <i>B.Sc in Mathematics, Computer Science, Physics</i>	2021 – 2024

EXAMS QUALIFIED

Qualified **UGC NET (Computer Science)**, June 2025 – Eligible for PhD admission.

PROJECTS

- **Movie Recommendation System** – Built an interactive Streamlit application that recommends movies based on user preferences using content-based filtering and cosine similarity. Utilized Pandas and Scikit-learn for data processing and model building, using the TMDB dataset for movie metadata.
[Live Demo](#)
- **Machine Learning Model for House Price Prediction** – Built a Scikit-learn pipeline with ColumnTransformer for preprocessing and trained a RandomForestRegressor using stratified sampling. Implemented model persistence and inference using Joblib for efficient predictions.
- **RAG-based Log Data QA App** – Developed a Streamlit-based Retrieval-Augmented Generation (RAG) application for natural language querying of log data. Integrated semantic retrieval and LLM-based generation to extract insights and patterns efficiently.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, SQL
- **Data Science & Generative AI:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI (LLMs, RAG), Data Visualization (Matplotlib, Seaborn), Pandas, NumPy, Scikit-learn, Streamlit
- **Frameworks:** TensorFlow, Flask, LangChain
- **Databases:** MySQL, NoSQL (MongoDB)
- **Core Concepts:** Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Operating Systems, Database Management Systems (DBMS)
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook